

DEQEN, China

WMO: 564440

Lat: 28.4886N

Lon: 98.9069E

Elev: 3320

StdP: 67.30

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.0	(c) -6.9	(d) -18.6	(e) 1.1	(f) 1.4	(g) -16.8	(h) 1.3	(i) 0.7	(j) 6.6	(k) 4.7	(l) 5.7	(m) 4.0	(n) 2.4	(o) 270	(p) 0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.5	(c) 21.3	(d) 12.5	(e) 20.1	(f) 12.3	(g) 19.1	(h) 12.0	(i) 13.8	(j) 18.6	(k) 13.4	(l) 18.0	(m) 13.0	(n) 17.3	(o) 2.5	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.4	13.6	14.7	12.0	13.3	14.4	11.7	12.9	14.3	51.4	18.9	50.2	18.3	48.9	17.4	19.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.2	4.4	3.7	-10.6	24.3	1.6	1.5	-11.7	25.4	-12.6	26.2	-13.5	27.1	-14.6	28.2
			-12.2	14.9	1.3	1.1	-13.1	15.7	-13.9	16.3	-14.6	16.9	-15.6	17.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-0.7	0.8	3.0	6.2	10.2	13.9	14.4	14.3	12.8	8.4	3.8	0.4
	DBStd	5.93	2.26	2.32	2.17	2.33	2.42	1.83	1.80	1.54	2.03	2.34	1.96	1.96
	HDD10.0	1502	332	258	216	116	27	1	1	0	4	61	187	298
	HDD18.3	4022	591	491	474	364	252	134	121	127	166	308	437	557
	CDD10.0	522	0	0	0	1	34	116	138	132	88	12	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	4	0	0	0	0	1	2	1	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
	Wind	WSAvg	1.8	2.0	2.0	1.9	1.9	1.6	1.3	1.4	1.5	1.8	2.0	2.0
(15) Precipitation	PrecAvg	725	15	26	66	72	67	75	143	119	70	49	17	5
	PrecMax	1018	55	73	190	233	162	140	243	179	166	145	104	25
	PrecMin	528	0	5	15	8	11	40	62	60	13	7	0	0
	PrecStd	127	13	20	44	49	40	26	38	35	36	31	22	6
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.5	12.2	13.8	17.0	20.9	23.1	22.5	22.5	21.2	18.4	13.0	9.9
		MCWB	0.7	2.1	3.9	6.9	9.7	12.7	13.3	13.0	12.2	9.7	3.8	0.7
	2%	DB	8.2	9.9	11.7	15.1	19.0	21.1	20.9	20.5	19.3	16.4	11.6	8.2
		MCWB	-0.2	1.2	3.1	6.0	9.4	12.1	13.2	12.9	11.7	8.7	3.3	0.1
	5%	DB	6.3	8.0	10.0	13.4	17.6	19.7	19.6	19.2	18.0	14.8	10.4	6.8
		MCWB	-1.3	0.5	2.4	5.4	8.9	11.8	13.1	12.8	11.4	7.8	2.9	-0.5
	10%	DB	4.6	6.3	8.3	11.8	16.0	18.4	18.3	17.9	16.8	13.4	9.0	5.4
		MCWB	-1.9	-0.2	1.9	4.9	8.4	11.7	12.9	12.6	11.2	7.0	2.2	-1.1
	0.4%	WB	1.8	3.1	5.3	8.0	11.6	13.9	14.5	14.2	13.4	10.8	6.0	2.0
		MCDB	8.6	9.0	10.7	15.2	17.6	19.3	20.0	19.2	18.4	15.7	9.3	8.0
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	0.3	1.9	4.1	7.0	10.6	13.4	14.0	13.7	12.7	9.9	4.9	0.8
		MCDB	6.7	7.8	9.0	13.2	16.1	18.5	18.9	18.3	17.3	13.8	8.6	6.4
	5%	WB	-0.7	1.0	3.3	6.2	9.9	12.9	13.6	13.2	12.1	9.2	3.9	0.0
		MCDB	4.9	6.7	8.1	11.5	15.3	17.8	18.0	17.5	16.4	12.7	8.6	5.5
	10%	WB	-1.5	0.2	2.5	5.5	9.2	12.4	13.1	12.8	11.6	8.5	3.0	-0.7
		MCDB	3.6	5.3	6.9	10.2	14.0	16.8	17.1	16.7	15.5	11.7	7.9	4.5
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	10.2	9.3	9.4	9.6	8.6	7.5	7.8	7.8	9.6	10.5	10.8
		MCDBR	13.0	13.9	12.9	12.8	13.0	11.3	9.9	10.1	9.9	11.4	12.4	12.3
		MCWBR	6.9	7.1	6.0	5.4	4.8	3.9	3.5	3.6	3.6	4.7	5.9	6.4
	5% WB	MCDBR	11.5	12.0	10.4	10.8	10.1	8.6	8.0	8.2	8.1	8.5	10.0	11.1
		MCWBR	6.4	6.3	5.3	5.0	4.0	3.4	3.1	3.3	3.2	3.7	5.2	6.0
(40) Clear-Sky Solar Irradiance	taub		0.242	0.279	0.348	0.346	0.312	0.300	0.307	0.316	0.292	0.259	0.234	0.222
	taud		2.555	2.445	2.234	2.321	2.517	2.650	2.634	2.572	2.631	2.711	2.707	2.678
	Ebn at Noon		1022	1000	939	948	975	980	972	966	988	1015	1031	1040
	Edh at Noon		84	103	137	130	107	93	95	100	91	78	72	71
(44) All-Sky Solar Radiation	RadAvg		4.01	4.44	4.67	4.81	5.24	5.17	4.74	4.68	4.39	4.37	4.17	3.98
	RadStd		0.44	0.37	0.57	0.53	0.41	0.40	0.41	0.44	0.28	0.36	0.34	0.25

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.27	N/A	N/A	+0.50	N/A	N/A	-89	-173	+73
(47) Regional (2 neighbors)		+0.44	N/A	N/A	+0.77	N/A	N/A	-74	-162	+104

Nomenclature: See separate page